

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA5112

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

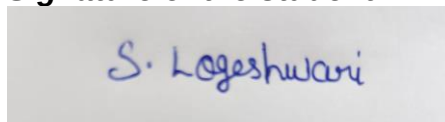
Annexure A

Code of Conduct:

I Logeshwari S (192565040) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink that reads "S. Logeshwari".

Annexure A Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$$p = 17, q = 19$$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

Given:

Prime number, **$p = 17$**

Prime number, **$q = 19$**

Public exponent, **$e = 5$**

Message, **$M = 7$**

Step 1: Calculate the Modulus (n)

The modulus is calculated using the formula:

$$n = p \times q$$

$$n = 17 \times 19$$

$$\mathbf{n = 323}$$

Step 2: Calculate Euler's Totient Function

The Euler Totient Function is calculated as:

$$\phi(n) = (p - 1)(q - 1)$$

$$\phi(n) = (17 - 1) (19 - 1)$$

$$\phi(n) = 16 \times 18$$

$$\phi(n) = 288$$

Step 3: Find the Private Key (d)

The private key satisfies the equation:

$$e \times d \equiv 1 \pmod{\phi(n)}$$

$$5 \times d \equiv 1 \pmod{288}$$

The value of **d = 173**

RSA Keys

Public Key = (5, 323)

Private Key = (173, 323)

Step 4: Encrypt the Message

Encryption Formula:

$$C = M^e \pmod{n}$$

$$C = 7^5 \pmod{323}$$

$$7^5 = 16807$$

$$16807 \pmod{323} = \mathbf{11}$$

Therefore,

Ciphertext (C) = 11

Step 5: Decrypt the Ciphertext

Decryption Formula:

$$M = C^d \bmod n$$

$$M = 11^{173} \bmod 323$$

$$M = 7$$

Therefore,

Recovered Message = 7

Security Analysis

The RSA algorithm is secure because it is based on the difficulty of factoring the large number **n** into its prime factors **p** and **q**. In this example, the prime numbers are very small (17 and 19), making it easy for an attacker to factor **323** and determine the private key.

In real-world applications, RSA uses prime numbers that are at least **2048 bits** long. Such large prime numbers make factorization computationally infeasible, thereby providing strong security against attacks.

Result

Modulus (n) = **323**

Euler Totient Function $\phi(n)$ = **288**

Public Key = **(5, 323)**

Private Key = **(173, 323)**

Ciphertext = **11**

Recovered Message = **7**

Thus, the RSA encryption and decryption process is successfully completed, and the original message is recovered correctly. The security of RSA depends on the use of very large prime numbers, which make it resistant to factorization attacks.

2. ElGamal Encryption Analysis

Given:

$p = 23$, $g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$

Compute the public key

Perform encryption to generate ciphertext pair (C_1, C_2)

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

Given:

Prime number, $p = 23$

Generator, $g = 5$

Private key, $x = 7$

Random integer, $k = 3$

Message, $M = 10$

Step 1: Compute the Public Key

The public key is calculated using the formula:

$$y = g^x \bmod p$$

$$y = 5^7 \bmod 23$$

$$5^7 = 78125$$

$$78125 \bmod 23 = 17$$

Therefore,

Public Key = (23, 5, 17)

Step 2: Encrypt the Message

First, calculate the first ciphertext component.

$$C_1 = g^k \bmod p$$

$$C_1 = 5^3 \bmod 23$$

$$C_1 = 125 \bmod 23$$

$$C_1 = 10$$

Next, calculate the second ciphertext component.

$$C_2 = M \times y^k \bmod p$$

$$C_2 = 10 \times 17^3 \bmod 23$$

$$17^3 = 4913$$

$$4913 \bmod 23 = 14$$

$$C_2 = 10 \times 14$$

$$C_2 = 140$$

$$140 \bmod 23 = 2$$

Therefore,

$$\text{Ciphertext} = (10, 2)$$

Step 3: Decrypt the Ciphertext

Using the private key, the receiver computes the original message from the ciphertext.

$$\text{Recovered Message} = 10$$

Thus, the decrypted message is the same as the original message.

Security Analysis

The El Gamal encryption algorithm uses a **random integer (k)** during every encryption process. Even if the same message is encrypted multiple times using the same public key, different values of **k** produce different ciphertexts.

This randomness provides the following security advantages:

- Prevents attackers from identifying repeated messages.

- Protects against replay attacks.
- Makes ciphertexts unpredictable.
- Improves the overall security of the encryption system.

Result

Public Key = **(23, 5, 17)**

Ciphertext = **(10, 2)**

Recovered Message = **10**

Thus, the ElGamal encryption and decryption process is completed successfully. The use of a random value **k** makes ElGamal more secure by generating different ciphertexts for the same plaintext, thereby improving resistance against cryptographic attacks.

3. RSA Digital Signature Verification

Given:

$p = 11$, $q = 17$, $e = 7$, message $M = 8$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

Given:

Prime number, **$p = 11$**

Prime number, **$q = 17$**

Public exponent, **$e = 7$**

Message, **$M = 8$**

Step 1: Calculate the Modulus (n)

The modulus is calculated using the formula:

$$n = p \times q$$

$$n = 11 \times 17$$

$$\mathbf{n = 187}$$

Step 2: Calculate Euler's Totient Function

The Euler Totient Function is calculated as:

$$\phi(n) = (p - 1)(q - 1)$$

$$\phi(n) = (11 - 1)(17 - 1)$$

$$\phi(n) = 10 \times 16$$

$$\mathbf{\phi(n) = 160}$$

Step 3: Find the Private Key (d)

The private key satisfies the equation:

$$e \times d \equiv 1 \pmod{\phi(n)}$$

$$7 \times d \equiv 1 \pmod{160}$$

The value of **$d = 23$**

RSA Keys

Public Key = (7, 187)

Private Key = (23, 187)

Step 4: Generate the Digital Signature

The digital signature is generated using the sender's private key.

Signature Formula:

$$S = M^d \pmod{n}$$

$$S = 8^{23} \pmod{187}$$

$$\mathbf{S = 57}$$

Therefore,

Digital Signature = 57

Step 5: Verify the Digital Signature

The receiver verifies the signature using the sender's public key.

Verification Formula:

$$M = S^e \bmod n$$

$$M = 57^7 \bmod 187$$

$$M = 8$$

The recovered message is equal to the original message.

Therefore, the digital signature is **valid**.

Non-Repudiation Analysis

RSA Digital Signature provides **Authentication**, **Integrity**, and **Non-Repudiation**.

Authentication: The receiver can verify that the message was sent by the legitimate sender because only the sender possesses the private key used to generate the signature.

Integrity: If the message is modified during transmission, the verification process will fail, indicating that the message has been altered.

Non-Repudiation: Since only the sender owns the private key, the sender cannot deny having signed the message. This ensures proof of origin and accountability.

Result

Modulus (n) = **187**

Euler Totient Function $\phi(n)$ = **160**

Public Key = **(7, 187)**

Private Key = **(23, 187)**

Digital Signature = **57**

Verified Message = **8**

Thus, the RSA Digital Signature is generated and verified successfully. It guarantees message authenticity, integrity, and non-repudiation, making RSA Digital Signature an essential technique for secure digital communication.

4. Diffie-Hellman with Attack Resistance Evaluation

Given:

$p = 29$, $g = 2$

User A private key = 5

User B private key = 12

Compute public keys for both users

Compute the shared secret key

Analyze what happens if an attacker intercepts and replaces public keys

Suggest a cryptographic solution to ensure authenticity

Given:

Prime number, **$p = 29$**

Generator, **$g = 2$**

User A's private key, **$a = 5$**

User B's private key, **$b = 12$**

Step 1: Compute the Public Keys

For User A,

$$A = g^a \bmod p$$

$$A = 2^5 \bmod 29$$

$$A = 32 \bmod 29$$

$$\mathbf{A = 3}$$

Therefore, **User A's Public Key = 3**

For User B,

$$B = g^b \bmod p$$

$$B = 2^{12} \bmod 29$$

$$B = 4096 \bmod 29$$

$$\mathbf{B = 7}$$

Therefore, **User B's Public Key = 7**

Step 2: Compute the Shared Secret Key

User A computes,

$$K = B^a \bmod p$$

$$K = 7^5 \bmod 29$$

$$K = 16$$

User B computes,

$$K = A^b \bmod p$$

$$K = 3^{12} \bmod 29$$

$$K = 16$$

Therefore,

Shared Secret Key = 16

Attack Resistance Evaluation

If an attacker intercepts the public keys exchanged between User A and User B, the attacker can replace them with fake public keys. As a result, both users unknowingly establish different secret keys with the attacker instead of with each other. This type of attack is called the **Man-in-the-Middle (MITM) Attack**.

Cryptographic Solution

The MITM attack can be prevented by using the following techniques:

- Digital Signatures
- Digital Certificates (Public Key Infrastructure - PKI)

These techniques verify the identity of the communicating users before establishing the shared secret key.

Result

Public Key of User A = 3

Public Key of User B = 7

Shared Secret Key = **16**

The Diffie–Hellman algorithm securely generates a shared secret key, but authentication techniques such as Digital Signatures and Certificates are required to prevent Man-in-the-Middle attacks.

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient

Justify its use in modern secure communication systems

Given:

RSA Key Exchange Time = **5 ms**

AES Encryption Time = **0.5 ms per data block**

Number of Data Blocks = **500**

Step 1: Calculate AES Encryption Time

AES Encryption Time

= Number of Blocks $\times$ Time per Block

= 500×0.5

= **250 ms**

Step 2: Calculate Total Hybrid Encryption Time

Total Time

= RSA Key Exchange Time + AES Encryption Time

= $5 + 250$

= **255 ms**

Therefore,

Total Hybrid Encryption Time = 255 ms

Step 3: Compare with RSA-only Encryption

If RSA is used to encrypt all 500 blocks,

RSA Time

$$= 500 \times 5$$

$$= \mathbf{2500\ ms}$$

Therefore,

RSA-only Encryption Time = **2500 ms**

Hybrid Encryption Time = **255 ms**

Performance Comparison

Hybrid Encryption is much faster because:

- RSA is computationally expensive and is mainly used for secure key exchange.
- AES is a fast symmetric encryption algorithm that efficiently encrypts large amounts of data.
- Combining RSA and AES provides both high security and high performance.

Real-World Applications

Hybrid encryption is widely used in:

- HTTPS (SSL/TLS)
- Online Banking
- Secure Email
- VPN (Virtual Private Network)
- Cloud Storage
- E-commerce Transactions

Result

AES Encryption Time = **250 ms**

RSA Key Exchange Time = **5 ms**

Total Hybrid Encryption Time = **255 ms**

RSA-only Encryption Time = **2500 ms**

Therefore, the Hybrid Encryption System is nearly **10 times faster** than using RSA alone while maintaining strong security. It is the preferred encryption method used in modern communication systems.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation